

Start time: 2:49 pm

Alpha and Beta Diversity

- Perform adjusted p-value!

Differential Abundance and Indicator Species Analysis

- Maybe filter by log2 fold change being bigger
- If the IN stage is the most consistently relevant, maybe write the paper on when the patients are within this stage the microbiota has specific biomarkers depending on sex for making predictions
- Compare between male vs. female → one graph for each disease stage → only 2 genus → indicator genus is not a good indication of distinction/predictor of gastric disease stage between sex

Functional analysis

- Can put the extra by group comparison in the supplementary files
- Do heatmap and the other analysis
- Filter log2fold change to 1 for IN (maybe)
 - `ggplot.geom(x= pathway names, y = log2fold change, size of bullet = p_adjust value)`
- Can fix the metabolite names using affinity later if they dont fit in

Overall

- Will probably have the paper focusing on specific stages that we found significant
- Better to separate graphs by stage at the moment
- Write from the perspective of a specific stage if we find a specific one to be significant
- Important to know for supplementary (?) whether the functionality of the microbiota is changing between the different sexes by itself or within the context of sex and gastric cancer progression
- For figures in final manuscript export images as PDF and modify in Affinity to clean up

Group Assignment 2

- Merge Aim 1 and 2 together
- Taxonomic differences (Aim 3 + 4) could arguably be one aim together
- Suggested to add another aim
 - Test random forest to see if it can predict sex and stage (potentially for only the IN stage, and test if the differences can predict the stage) ***likely cannot do RF due to lack of multiple indicator species***
- Parametric test for alpha diversity
- Ensure to be clear and justify what types of comparison you are doing for the proposal

Proposal

- Write the proposal as if someone has never read it before (?)